Akash Prajapati

Python Backend Engineer | Applied AI & LLM Systems

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PROFESSIONAL SUMMARY

Python Backend Engineer with 3.4+ years of experience developing scalable backend systems, REST APIs, and microservices using Python, Django, DRF, and FastAPI. Experienced in database optimization, asynchronous processing, and enterprise integrations, with hands-on exposure to Applied AI, RAG pipelines, LLM integrations, and semantic search across automotive, e-commerce, and AI-driven applications.

TECHNICAL SKILLS

Programming & Backend: Python, SQL, Django, Django REST Framework, FastAPI, REST APIs, Microservices

Applied AI & GenAI: LLMs, RAG, Prompt Engineering, LangChain, LangGraph, Tool Calling, OpenAI APIs, Semantic Retrieval, Image Embeddings, Sentence Transformers, ChromaDB, Whisper, ElevenLabs, Guardrails, LLM Harnessing, LLM Evaluation, LangSmith

Databases & Messaging: PostgreSQL, Redis, Django ORM, Celery, Apache Kafka

Authentication & Integrations: Keycloak, JWT, OAuth 2.0, OTP Verification, Payment Gateway Integration, Third-Party API Integration, RBAC

Concepts: OOP, Database Design, Query Optimization, Agile Methodology, System Design

Cloud & Tools: AWS (S3, RDS, EC2, ELB), Docker, Kubernetes (Fundamentals), Grafana, Git, GitHub, Swagger/OpenAPI

PROFESSIONAL EXPERIENCE

Vishleshan Software Solutions Pvt. Ltd. | Software Developer 25th November 2025 – 24th September 2026

- Developed enterprise e-commerce backend workflows using Python, Django, DRF and PostgreSQL (AWS RDS), contributing to 100+ REST APIs supporting 1,000+ daily orders, with integrations for DSR/MSR operations, Shiprocket shipment tracking and SAP Grow.
- Developed FastAPI-based notification services integrated with Apache Kafka, supporting approximately 1,000–10,000 notifications daily through event-driven processing.
- Implemented Redis caching for Order History APIs and optimized Django ORM queries to reduce repeated database access and improve API response times.
- Built asynchronous report generation, export and scheduled processing workflows using Celery and Redis, integrating AWS S3 for file storage and secure presigned downloads.
- Implemented product discovery using separate image and text embedding pipelines, supporting image, part-number, product-name, category and voice-based search with top-five similarity results.
- Developed backend APIs and an end-to-end RAG pipeline using Python and Django, implementing document chunking, Sentence Transformers embeddings, ChromaDB storage and semantic retrieval for a Hindi voice-enabled assistant.
- Integrated Whisper, GPT-4 and ElevenLabs APIs for speech-to-text, response generation and text-to-speech, enabling a Hindi voice interaction workflow.
- Evaluated approximately 5,000 test cases against expected answers using LLM-based evaluation, achieving an average reported evaluation score of approximately 87%.
- Integrated Keycloak authentication APIs using JWT and OAuth 2.0, implemented CAPTCHA validation and supported backend maintenance and production issue resolution.

Sankey Solutions | Solution Analyst 1st June 2023 – 14th November 2025

- Developed backend features across vehicle, wrapper, subscription and payment services within an approximately 18-microservice fleet management platform using Python, Django and PostgreSQL.
- Built a vehicle day-wise tracker backend end-to-end, processing historical GPS coordinates, calculating distances using the Haversine formula, identifying trips and stops, and preparing route data for Google Maps visualization.
- Improved API performance by 20–30% by identifying and resolving N+1 query issues and optimizing database access using Django ORM Subquery and related query optimizations.
- Implemented role- and subscription-based access controls, managing Fleet Owner and Fleet Manager permissions, site-level vehicle visibility and plan-based access to vehicle data and fleet management modules.
- Developed mileage analytics APIs and business logic to process speed, idling, harsh acceleration, braking and gear shifting/RPM parameters to generate mileage-improvement guidance.
- Implemented historical telemetry synchronization through third-party APIs using 30-minute cron scheduling, checkpoint-based incremental fetching, backlog recovery and PostgreSQL upserts keyed by VIN and timestamp.
- Integrated CCAvenue payment gateway with request encryption, response decryption, payment callbacks, status updates, subscription synchronization and Redis cache updates; developed workflows for subscription purchases, upgrades, renewals and expiry.
- Designed and developed a quality-management backend using Python, Django and PostgreSQL, including database design and 50+ REST APIs to digitize vehicle assembly-line inspections, defect reporting and corrective-action workflows.

- Implemented Quality Maker and Checker workflows and developed an Apache Kafka consumer to receive VIN and vehicle information from manufacturing sites and persist incoming data in PostgreSQL; integrated Swagger/OpenAPI documentation.
- Independently developed an AI-assisted Git repository analysis solution integrating GitHub APIs and OpenAI APIs for commit-change summarization and potential security issue detection, generating structured reports identifying exposed API keys, access tokens, passwords, email credentials and AWS access details.

EDUCATION

Bachelor of Engineering — Computer Engineering
Smt. Indira Gandhi Engineering College, University of Mumbai | Aug 2019 – Jun 2023 | CGPA: 8.88/10

ACHIEVEMENTS

- **Shining Star:** Recognized for outstanding technical contributions.
- **Bar Raiser:** Recognized for problem-solving, technical guidance, and supporting team members.

CERTIFICATIONS

- Machine Learning with Google Cloud
- Android Development

PUBLICATION

- Crop Protection Against Birds Using Deep Learning and IoT, IRJET, Vol. 10, Issue 4, April 2023.